

Experimentally assessed efficiency improvement of innovative 3D-shaped structures as volumetric absorbers

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ABSTRACT

Solar tower plants are predicted to take on a significant role in the ongoing global energy transition. The open volumetric air receiver technology can provide an alternative to molten salt receivers due to its proven and effective storage capabilities. The key component of this technology are porous absorber structures which are responsible for the absorption of the solar irradiation. Research of porous structures has consistently shown that 3D-shaped absorber designs offer the highest efficiency potential. The realization of such designs is made possible by newer manufacturing techniques.

In this publication, experimental results for two groups of new volumetric absorbers are presented. The StepRec absorber is manufactured via ceramic 3D screen printing, its characteristic step-pin sections allow for deep penetration of the solar irradiation. Additionally, metallic foil absorbers were evaluated, they exhibit particularly high specific surface areas due to low wall thicknesses. The new absorbers were tested for flux densities up to $650 \frac{\text{kW}}{\text{m}^2}$ and air exit temperatures up to 735°C . The experiments successfully confirmed the new absorbers' great potential: They outperformed the HiTRec reference, a ceramic honeycomb structure, for all air temperatures. At 650°C , thermal efficiencies up to 91.8% were recorded, an improvement of 6.5%-points w.r.t. to the HiTRec result.

1. Introduction

CSP and especially solar power tower plants are expected to play an important role in the ongoing global energy transition [1]. With its proven and effective storage, the open volumetric air receiver is a real alternative to the currently prevailing molten salt receiver technology, as was already stated 1993 in the Second Generation Study [2]. Advantages of the air receiver technology lie in the simplicity of design, the robustness of operation and the virtually unlimited temperature range of air as heat transfer medium. Therefore, since many years, continuing effort has been put into the development of the key component of this technology, porous structures as volumetric absorbers of concentrated sunlight. A comprehensive overview of this technology development can be found in the detailed review by Ávila-Marín [3].

1.1. First generation of absorber structures

Volumetric absorbers allow the radiation to penetrate deep into their volume to be absorbed and transferred as heat to the parallel gas flow by the large inner surface. Unsurprisingly, woven or knitted meshes of thin metal wires from temperature resistant steel alloys dominated the first steps of this development [3]. Fricker [4] shall be mentioned here, who pioneered with a seven-layer wire mesh absorber made of 0.71 mm heat-resisting CrNi steel wire in 1985 which lead to the $2.5 \text{ MW}_{\text{th}}$ TSA receiver tested in the Spanish Plataforma Solar (PSA) in 1993, composed of hexagonal absorber modules packed with rolled layers of knitted Inconel® wire [5].

Further structures examined in the early phase of development were either regular channel-like geometries or isotropic reticulated

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Nomenclature

Acronyms

CFD	Computational Fluid Dynamics
CSP	Concentrated Solar Power
DLR	German Aerospace Center
HiTRec	High Temperature Receiver
OVR	Open Volumetric Receiver
reSiC	reaction-bound silicon carbide
SiSiC	siliconized silicon carbide

Greek Symbols

α	Absorptivity (–)
δ	Stochastic uncertainty variable (unit varies)
η	Efficiency (–)
λ	Wavelength (m)
μ	Mean/expected value (unit varies)
ρ	Reflectivity (–)
σ^2	Variance (unit varies)

Roman Symbols

c	Coefficient (–)
d	Direction vector (–)
I	Incident irradiance ($\frac{W}{m^2}$)
n	Normal vector (–)
n	Number/quantity (–)
p	Position vector (–)
$\dot{q}$	Heat flux ($\frac{W}{m^2}$)
T	Temperature (K)
ti	Tilt angle (°)
u	Uncertainty (unit varies)
v	Rel. measurement value (–)
w	Weight (–)

Subscripts

abs	Absorption
adj	Adjusted
air	Air
amb	Ambient
avg	Average
BL	Base-line
dir	Direction
emi	Emission
IR	Infrared
j/k/l	Running index
loss	Loss
meas.	Measured
pos	Position
rand	Random
rect	Rectified
refl	Reflection
s.w.	Solar-weighted
sim.	Simulated
sol	Solar
th	Thermal
TR	Test rig
ZL	Zero-line

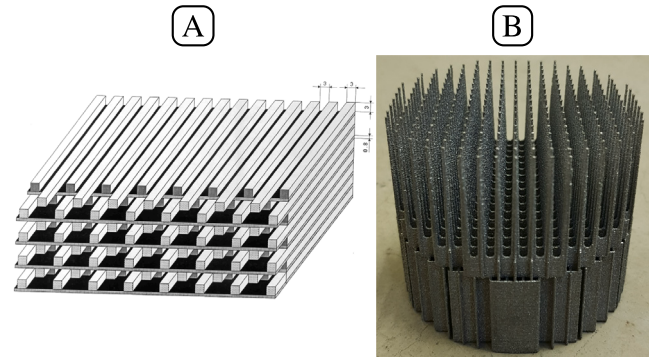


Fig. 1. Foil-rod absorber by DLR and Hoechst CeramTech (image A, image source: [9], units given in millimeters) and the titanium–aluminum demonstrator absorber as described in Capuano et al. [10] (image B, image source: DLR).

foam structures [6,7]. Channel-like absorbers were made from metal by stacking alternately corrugated and non-corrugated metal foils, similar to catalyst carriers deployed in the automotive sector [7]. These absorbers feature extremely fine wall structures (50 μm) and a high open porosity (cf. Section 2.2). The proposed ceramic channel structures were either composed of extruded ceramic foils and rods or directly fabricated as extruded channel monoliths [7]. Silicon Carbide ceramics of different composition and structure (reSiC, SiSiC) was (and is) the prevailing material used. While ceramic material naturally exhibits excellent temperature resistance, production by extrusion allows only comparatively coarse structures with wall thicknesses in the range 0.4 – 0.8 mm. Ceramic foam structures are usually produced by molding an open-pore polymer foam with ceramic slurry. The polymer is completely vaporized during the firing process, resulting in a hollow core for the struts [6,8].

1.2. Variation of structural geometry along the flow direction

The structures of this first phase of development have in common that they almost exclusively exhibit uniform geometric parameters throughout the volume, i.e. single wall size, channel width, mesh size etc. But, some first attempts can be found in literature to alter the design of the frontal structure, with the purpose to change geometric, optical or heat-transfer related parameters in flow direction.

Bechtel tested a wire mesh absorber composed of several layers of thin 0.1 mm wire followed by layers with thicker 0.2 mm wires [11]. DLR and Hoechst CeramTec developed and tested a ceramic channel structure composed of foils and rods where in the front section every second foil and every second rod were shortened [9] (cf. Fig. 1). Fend et al. [12] tested the combination of a thin front layer of an 80ppi ceramic foam covering a thicker main absorber of 20ppi foam of the same material. All these attempts tried to reduce thermal losses by offering a higher porosity at the front resulting in deeper penetration of the radiation into the volume of the structure.

After these early, rather exploratory design variations, attempts were made to vary the geometry of porous absorber structures more purposefully. Roldán et al. [13] used a 2-dimensional CFD-model for the comparison of ceramic honeycombs of varying porosity in radial and axial direction with reference to a structure of constant porosity, finding decreasing porosity in flow direction to be the best performing structure. Ávila-Marín et al. [14,15] experimentally investigated the performance of gradual porosity absorbers consisting of combinations of wire meshes with different geometrical properties, concluding that double layer meshes with the front layer having the highest possible porosity and the second layer having the highest possible specific surface area give the best results. Zaversky et al. [16] numerically and experimentally evaluated single and multi-layer ceramic foams as solar

absorbers and found out that the front structure should have the highest possible porosity and that multi-layer structures show no significant advantage in efficiency. Finally, Capuano et al. [17] proposed an innovative channel structure with a gradual decrease of porosity achieved with deep cuts in the channel walls. An additionally high inner surface area was realized through a staggered second channel layer. Due to its delicate structure, this absorber could not be realized from the usual target material for solar application (ceramics or temperature resistant steel alloys) but had to be made as a demonstrator from a titanium–aluminum alloy through additive manufacturing in a scale of 3:1 to the original proposed geometry [10] (cf. Fig. 1). Nevertheless, this demonstrator could be tested under irradiation of $760 \frac{\text{kW}}{\text{m}^2}$ up to a temperature of 590°C , showing a thermal efficiency of about 82%.

1.3. Structural design adapted to manufacturing processes

This work is based on the idea of incorporating the possibilities and limitations of the chosen manufacturing methods and materials into the design process. Therefore, specialists from the manufacturing companies were involved in the development of the specific structural designs. The work focused on two production technologies:

- innovative ceramic honeycomb structures manufactured by screen printing out of SiSiC
- channel structures made from pairs of planar and corrugated metal sheets, similar to those presented by Pabst et al. [18]

A new simulation model was applied as part of the design process. This model and related simulation results are described in detail in Broeske et al. [19]. From the most promising geometries, several design variations were produced and tested under close-to-real experimental conditions in DLR's large solar simulator Synlight® [20]. As a benchmark, the state of the art of the so-called HiTRec technology was used: an extruded honeycomb monolith made from SiSiC ceramics with a 2.0 mm channel width and a 0.8 mm wall thickness as it is deployed in the Solar Tower in Jülich (Germany), where the technology of the open volumetric receiver is demonstrated as a power plant system since 2009 [21].

In the next section the chosen manufacturing methods and the implications and limitations on the design process are presented. Afterwards, the test setup and the experimental parameters are described to assess the conditions of the measurement procedure. The experimental results are presented and discussed in the subsequent chapter. Finally, the findings are summarized and conclusions drawn.

2. Absorber structures

2.1. Screen printed ceramics

The first new absorber design is manufactured via a 3D screen printing process developed by the company Exentis Group AG [22]. As with 2D screen printing, 3D screen printing is based on constructing a stencil on a fine mesh screen and pushing a special paste through the printing screen to create an imprint of the design on the surface beneath. To extend this concept into 3D parts, this process is repeated continuously while the printing screen is moved upwards. This causes the printed part to “grow” layer by layer. Depending on paste and printing parameters, each layer is usually between $7 - 40 \mu\text{m}$ thick. For very precise and filigree parts, a small layer increment is preferred whereas larger increments lead to faster growing parts and therefore a decrease in process time. To change the geometry in the z-axis direction, a switch of the printing screen is necessary. In theory, using a new screen for each layer would be possible for maximum geometrical flexibility, but this is not practical when taking economic feasibility into account. Therefore, the volumetric receiver proposed by Capuano et al. [10] with sloped pins was adjusted to better fit the 3D screen printing

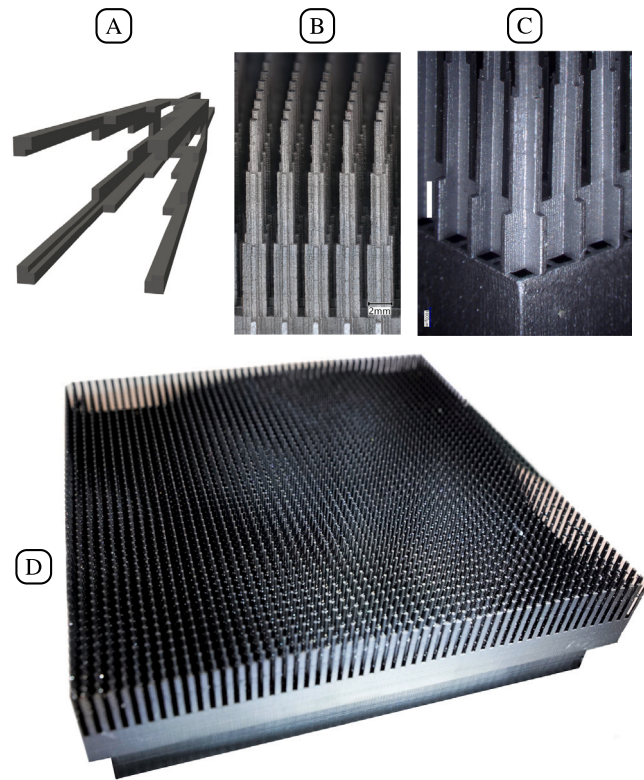


Fig. 2. The StepRec absorber design: geometric single channel model (image A) and 3D printed absorber sample (image D: full absorber matrix, images B + C: zoomed views).

technology: The sloped pins were replaced by three “step”-like sections (cf. Fig. 2) giving name to the so called StepRec design.

Despite the simple concept of 3D screen printing, the actual printing of 3D parts is quite complex. It requires highly developed machines, paste, screens and printing parameters, all of which must be fine-tuned to each other. The flexibility in material selection is possible due to applying pastes, which usually combine the desired material in powder form with binder and other additives. In case of the solar absorbers, a specialized paste containing silicon carbide and carbon powder as the base material, binders to realize sufficient green body stability, a fluid carrier medium and rheology additives was developed. The additives serve to optimize the rheological behavior of the 3D screen printing paste. Usually, a shear-thinning behavior is desirable to prevent unwanted seep when the sieve is flooded with paste, while still allowing for smooth flow when moved by the squeegee.

The StepRec design consists of five different layers, two for the base with offset square channels and the three step-like pin layers. The channels itself are no challenge to print. A challenge arises from the fact that the channels are offset on top of each other, since a new layer can only be built on an underlying foundation. Depending on the existing structure and gap width, it is possible to build an arched bridge, because the paste sticks to the before printed structures and narrows the gap with each layer. Unfortunately, this often reduces the stability of the green body and therefore can lead to problems with increasing gap width and the height on top. The better solution in this case was printing layer one and layer two to five separately and then join them together with a small amount of paste as adhesive that was printed on layer one.

After being printed, the absorber samples are sintered in a combined thermal process [22]. During this process, the organic binder materials are removed, before the solar absorber is infiltrated with molten silicon. The additional silicon fills remaining pore structures within the absorber material and partially reacts with free carbon to secondary SiC.

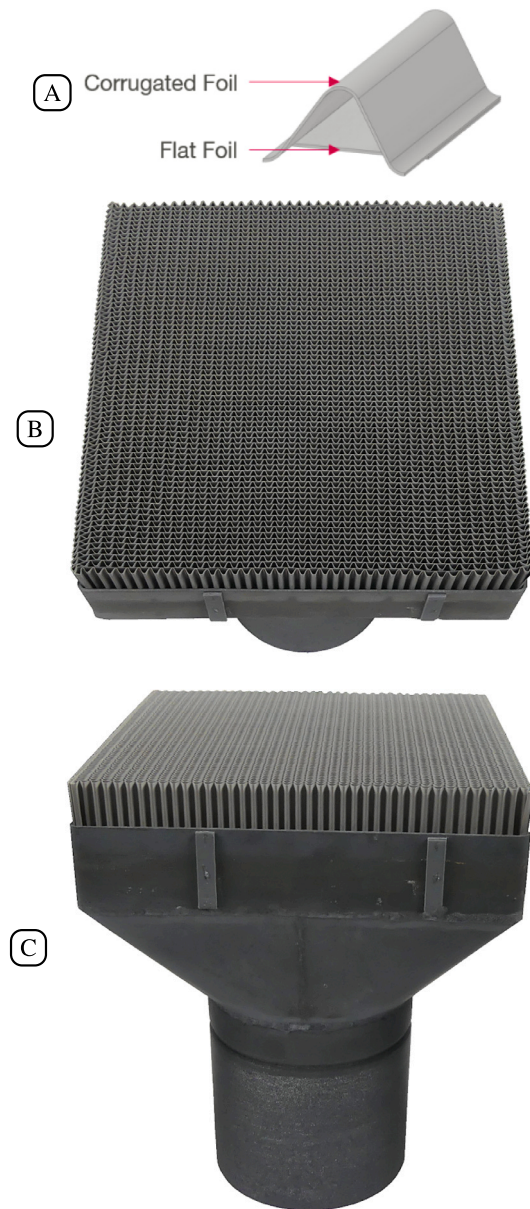


Fig. 3. The EmiRec absorber design: geometric channel model (image A) and a full-size absorber module (images B + C).

This reaction bonded silicon carbide allows for very low porosity with almost no shrinkage during thermal processing and therefore reduces warpage and inner strain compared to sintered silicon carbide.

2.2. Soldered metal sheets

The evaluated metallic solar absorbers were manufactured by the company Vitesco Technologies Emitec, utilizing the company's long-standing experience and know-how from the automotive exhaust gas aftertreatment industry. These absorbers, hereafter referred to as EmiRec absorbers, are made of a modified high-temperature iron-based alloy specially developed for corrosive high-temperature applications (1.4767 MOD - iron-chromium-aluminum alloy). The material has the property of forming a protective aluminum oxide layer during operation at high temperatures, which prevents oxidation of the inner material core.

The matrix structure of the EmiRec absorbers consists of thin cut-to-length sheet foils, which are stacked alternately from a corrugated

layer and a smooth layer to form a package and inserted into a square tube (cf. Fig. 3). At defined contact points between the sheet foils and between the matrix package and the square tube, a material-to-material bond is produced by means of the vacuum brazing technology. The connections between the sheet foils and between the matrix package and the tube provide mechanical strength while at the same time being flexible enough to compensate for stresses in the material and prevent plastic deformation under high thermal shock loads. High thermal shock loads occur especially during cloud passages when the heliostats are suddenly shaded.

The square design of the EmiRec solar absorbers makes optimum use of the available installation space in the receiver field on the solar tower and can be individually adapted to the boundary conditions of different solar power plants. The high-temperature material used is processed in thin sheet metal strips, which are characterized in particular by good heating behavior, low heat capacity and very good axial and radial thermal conductivity. The low wall thickness of the metal foils allows a high porosity despite a large specific surface in direct comparison with the state of the art SiSiC absorber. For the tests in the Synlight facility, the solar absorbers were pre-oxidized in a high-temperature furnace, which darkens the surface of the high-temperature material. The darkened surface improves the absorptivity and thus increases the thermal efficiency of the solar absorbers. During operation on a solar power plant, the reaction with oxygen takes place and forms the oxide layer after a short period of operation.

In upstream production tests, various design approaches for increasing the inlet porosity of the metallic EmiRec solar absorbers were investigated. Eventually, it was decided to shorten or retract the smooth layer and set it flush with the gas outlet side (cf. Fig. 3, image A). By shortening the smooth layer in the matrix package, the axial porosity can be adjusted in relation to the angle of incidence of the solar irradiation. As part of the Synlight test campaign, a number of designs were evaluated to test various manufacturing characteristics. However, this publication will focus on the three most promising designs: The first two, EmiRec 300 cpsi and EmiRec 600 cpsi, are standard designs with the smooth layers shortened by 10 mm and differ only regarding the channel density. For the third design, EmiRec 600 cpsi MOD, not only the smooth layers were shortened, but every second corrugated layer was also shortened by 10 mm. This change further increases the inlet porosity.

3. Experimental setup

3.1. Synlight test setup

The test rig was designed and built by the German company Kraftanlagen Energies and Services GmbH, images of the rig are shown in Fig. 4. It is based on the HiTREC open volumetric receiver technology, i.e. the same technology which forms the receiver of the Solar Tower Jülich power plant. A total of sixteen, full-sized (140 mm by 140 mm) absorber modules can be evaluated simultaneously. An air return system was not implemented. The test rig was also used in a previous test campaign where absorber samples with a reduced size (60 mm by 60 mm) were evaluated [23].

For the measurement of the air exit temperatures, the test rig was equipped with thermocouples placed directly behind each of the sixteen absorber (matrix) positions. Individual mass flow measurements for each absorber were impossible due to space constraints. The absorber efficiencies were therefore evaluated via a radiative energy balance, rather than via the air enthalpy flow. A similar approach has been previously used by Pabst et al. [18] to experimentally assess absorber efficiencies. During the experiments in the Synlight facility, thermal emissions of the absorber probes were recorded using an infrared camera, which was placed between the Synlight lamps viewing the test rig at an angle of around 20 degrees. From the infrared recordings, the thermal radiation losses per unit area $\dot{q}_{loss,emi}$ were derived for each

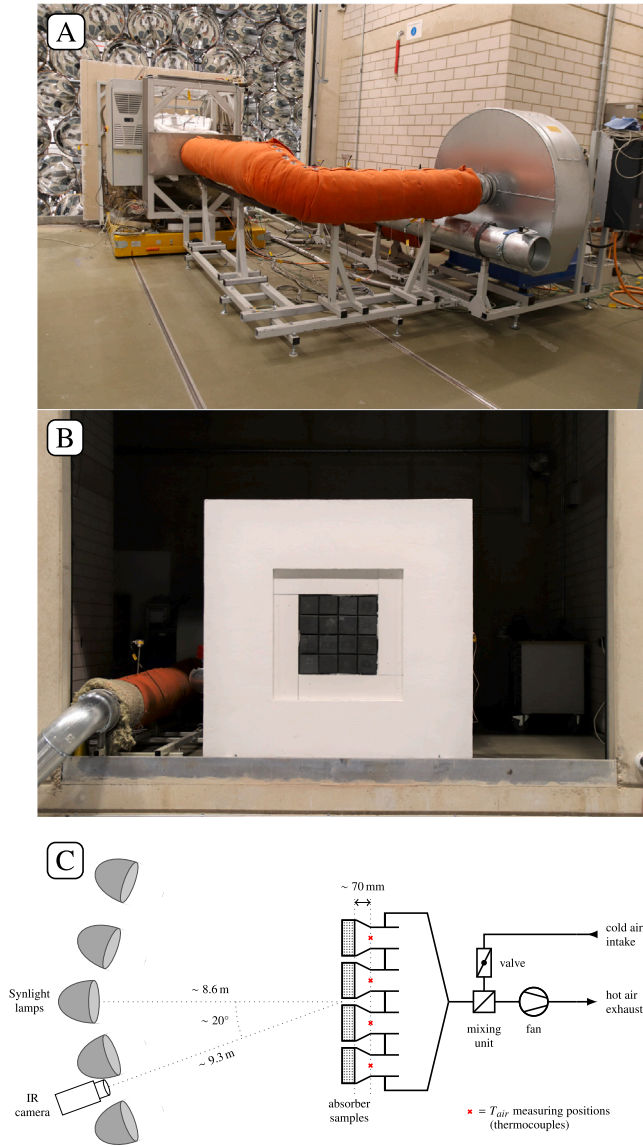


Fig. 4. Experimental test rig for the Synlight campaign. Shown are a rear view (image A), a front view with installed HiTRec 80cps absorbers (image B), and a schematic top-down view (image C, not to scale).

absorber probe and temperature level. The corresponding air exit temperature T_{air} was determined by taking the average of the thermocouple data around the IR-image time-stamps. The average incident flux density I_{avg} was measured with DLR's FMAS system [24]. The reflection losses of each absorber were computed from measurements with a spectrometer (cf. Section 3.2) leading to a solar-weighted absorptivity value $\alpha_{s,w}$. The absorber efficiency is then given by:

$$\eta_{th}(T_{air}) = \alpha_{s,w} - \frac{\dot{q}_{loss,emi}(T_{air})}{I_{avg}} \quad (1)$$

The thermal radiation measured on the absorber surface by means of an IR camera unfortunately also contains a certain amount of reflected IR radiation from the electric Synlight lamps. This is unintended as the reflection losses are already accounted for in the solar-weighted absorptivity. To compensate, these reflections $\dot{q}_{refl,IR}$ were estimated based on Eq. (2). The thermal efficiencies were extrapolated for air exit temperatures equal to the ambient temperature assuming that the efficiencies should asymptotically trend towards the solar-weighted

absorptivity. The observed differences were then interpreted as the unintentionally measured reflections and rectified thermal efficiencies were defined as shown in Eq. (4).

$$\dot{q}_{refl,IR} = I_{avg} \left(\alpha_{s,w} - \lim_{T \rightarrow T_{amb}} \eta_{th}(T) \right) \quad (2)$$

$$\dot{q}_{loss,emi,rect}(T_{air}) = \dot{q}_{loss,emi}(T_{air}) - \dot{q}_{refl,IR} \quad (3)$$

$$\begin{aligned} \eta_{th,rect}(T_{air}) &= \alpha_{s,w} - \frac{\dot{q}_{loss,emi,rect}(T_{air})}{I_{avg}} \\ &= \alpha_{s,w} - \frac{\dot{q}_{loss,emi}(T_{air}) - \dot{q}_{refl,IR}}{I_{avg}} \end{aligned} \quad (4)$$

3.2. Solar-weighted absorptivity measurements

The solar-weighted absorptivities were determined for each absorber sample based on separate measurements with a Perkin Elmer Lambda 950 spectrometer conducted at the DLR location in Cologne, Germany. The spectrometer measurement themselves first yielded spectral reflectivity values, which were then converted to the solar-weighted absorptivity. Data was recorded for wavelengths between 250 nm and 2500 nm with a 10 nm step size. For each absorber sample, three measurements were conducted aiming to reduce random errors, e.g. caused by the sample placement. To further improve the accuracy, wavelength-dependent data was also recorded for a reference material¹ with known spectral reflectivity values, called the “base-line”, and another run without any sample placed in the spectrometer yielding the “zero-line”. The spectral reflectivity for each wavelength λ of an absorber sample was then calculated as shown in Eq. (5) Here, $v_{avg,\lambda}$ represents the wavelength-dependent data average of the three sample measurements, $v_{ZL,\lambda}$ the zero-line data, $v_{BL,\lambda}$ the base-line data and $\rho_{BL,\lambda}$ the base-line spectral reflectivities. Finally, the solar-weighted absorptivity $\alpha_{s,w}$ of each sample is the weighted average of the spectral absorptivities, the spectral weights w_λ are defined based on a standard solar spectrum (E891):

$$\rho_\lambda = \rho_{BL,\lambda} \cdot \left(\frac{v_{avg,\lambda} - v_{ZL,\lambda}}{v_{BL,\lambda} - v_{ZL,\lambda}} \right) \quad (5)$$

$$\alpha_{s,w} = \sum_{\lambda=250 \text{ nm}}^{2500 \text{ nm}} w_\lambda (1 - \rho_\lambda) \quad (6)$$

3.3. Test procedure

The main test campaign was carried out at the high-flux solar simulator facility Synlight[®] of the DLR in Jülich, Germany. The core of the campaign then consisted of a series of test days, where data was recorded for one absorber configuration set and one flux density distribution during one test day (cf. Section 3.4).

The goal was to determine the thermal efficiency of the different absorbers at steady state operation. To achieve this, a series of different temperature target levels were evaluated over the course of one test day in a “pyramid”-pattern from low to high to low. The temperature levels were controlled via the total air mass flow through the test rig. The actual outlet temperatures varied between the sixteen absorber samples, in part because of their different efficiencies. Hence, the temperature levels served only as target values. For each level, the actual evaluation period started once the temperature gradients of all absorbers dropped and stayed below $2 \frac{\text{K}}{\text{min}}$ for ten minutes. The duration of the evaluation periods was then also set as ten minutes, during this time at least three infrared images were recorded. The air exit temperatures were recorded continuously by the test rig. In total, the highest recorded

¹ The used reference material is called “Spectralon” (LBS-SRS-05-020 (SRS 3644)) manufactured by the company Labsphere.

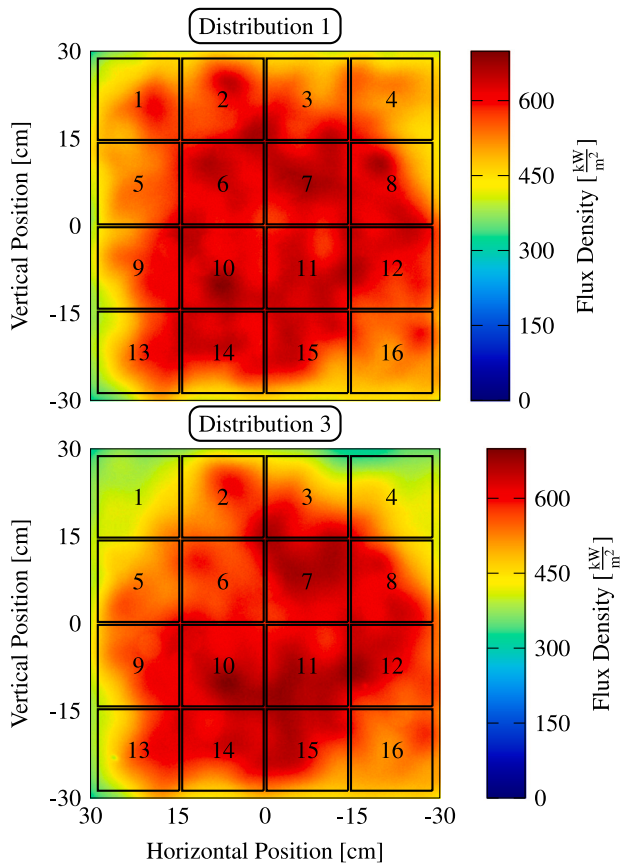


Fig. 5. Flux density measurement results for the first (“Distribution 1”, top) and the third (“Distribution 3”, bottom) distribution. Black squares and numbers indicate the sixteen absorber positions.

air exit temperatures were 715 °C for the ceramic StepRec samples and 725 °C for the metallic EmiRec absorbers. Higher temperatures were not possible due to the temperature limits of subsequent components of the test rig.

3.4. Absorber positioning and flux density measurements

During the test campaign, four different sets of absorber configurations were used. Each set consisted of sixteen absorber samples and each sample was assigned to one of the sixteen positions within the test rig. The metallic and ceramic absorbers were evaluated separately. The first set consisted of only HiTRec 80 cpsi reference samples, the second of the Exentis samples plus HiTRec samples to fill the remaining positions. The third and fourth absorber set then were fully composed of the metallic EmiRec geometries. The positions of the five absorber geometries whose results are presented in this publication are listed in Table 1. The numbering of the absorber positions is shown as part of the flux distribution visualization in Fig. 5.

Three different flux distributions were used during the experimental campaign, all with the same target level of $600 \frac{\text{kW}}{\text{m}^2}$ to $650 \frac{\text{kW}}{\text{m}^2}$. The first and third distribution are visualized in Fig. 5. The average flux density values for the positions of the selected absorbers designs are given in Table 2. For the configuration of the Synlight lamps’ alignment, the priority was placed on achieving a homogeneous flux distribution for the inner positions (6, 7, 10, 11). Distribution 2, which is not shown in Table 2 due to its similarity to distribution 1, was used as a temporary distribution for a single test day. Compared to distribution 1, the slightly modified distribution 2 allowed different absorbers to reach

Table 1

Set and position assignments for the selected 5 absorber geometries.

Absorber	Sample 1		Sample 2	
	Set	Pos.	Set	Pos.
HiTRec 80 cpsi			Set 1 + Pos. 1 – 16	
StepRec	2	6	2	10
EmiRec 300 cpsi	4	7	4	15
EmiRec 600 cpsi	3	7	4	14
EmiRec 600 cpsi MOD	3	10	4	12

Table 2

Average flux density for the three flux distributions and selected absorber positions.

Position		6	7	8	10
Distr. 1	$[\frac{\text{kW}}{\text{m}^2}]$	628.7	637.2	591.8	635.0
Distr. 2	$[\frac{\text{kW}}{\text{m}^2}]$	624.0	634.5	593.3	615.9
Distr. 3	$[\frac{\text{kW}}{\text{m}^2}]$	593.1	641.5	583.9	639.0
Position		11	12	14	15
Distr. 1	$[\frac{\text{kW}}{\text{m}^2}]$	632.2	611.8	597.7	586.5
Distr. 2	$[\frac{\text{kW}}{\text{m}^2}]$	618.0	604.9	505.0	530.7
Distr. 3	$[\frac{\text{kW}}{\text{m}^2}]$	644.0	613.3	602.3	605.5

a maximal air exit temperature. The third distribution had to be used during the latter portion of the test campaign due to the unavailability of some lamps. The absorber set 1 (Table 1) was evaluated with distribution 1, set 2 with distributions 1 and 2, and sets 3 and 4 with distribution 3.

3.5. Uncertainty analysis

The uncertainty of the conducted experiments has been analyzed based on the methodology described in the “GUM: Guide to the Expression of Uncertainty in Measurement” [25].

Based on the manufacturer specifications for the thermocouples (type N, TMH [26]), the maximum temperature uncertainty can be expected to be below ± 3 °C. Compared to the other sources of uncertainty, this is considered as negligible.

The general uncertainty of the FMAS measurement system under ideal conditions is up to three percent [24], a more conservative uncertainty δ_{FMAS} of up to six percent is assumed here based on experience from other DLR experiments. Deviations between the expected positioning and the actual positioning of the absorber samples within the test rig were considered as an additional uncertainty for the evaluation of the flux density I_{avg} and the emission losses $\dot{q}_{loss,emi}$. These uncertainties $\delta_{pos,sol}$ and $\delta_{pos,emi}$ were estimated via two sensitivity studies testing differences of up to 1 cm in any direction. Regarding the emission losses, the positioning of the IR camera relative to the test rig and the general accuracy of the IR camera were discussed as further sources of uncertainty, but their influence could not be quantified.

The evaluation process of the solar-weighted absorptivity is specifically designed to reduce random errors (cf. Section 3.2). As an estimator of the random errors (e.g. geometric sample impurities), the solar-weighted absorptivity evaluated based on the averaged measurements was compared to the absorptivity values calculated from each individual measurement. This comparison resulted in an uncertainty $\delta_{abs,rand}$ of up to one percentage point. Systemic deviations between the measurements with the spectrometer and the irradiation tests in the Synlight were considered w.r.t. the irradiation direction. The spectrometer measures the sample reflectivity for a shallow incident angle of eight to ten degrees. For the Synlight facility, higher incident angles are possible depending on which lamps are used. This uncertainty $\delta_{abs,dir}$ was quantified based on a comparison of raytracing simulations, this is explained in Appendix A. Deviations of the spectral light distribution between the Synlight lamps and spectrometer evaluation were considered, but deemed negligible as the Synlight facility was specifically designed to mimic real solar irradiation conditions.

Table 3

Solar-weighted absorptivity results for the different absorber samples. For the HiTRec 80 cpsi absorbers, the range for the samples placed on the inner positions (6, 7, 10, 11) is shown.

Absorber	Sample 1	Sample 2
HiTRec 80 cpsi	90.4%–91.1%	
StepRec	97.2%	97.2%
EmiRec 300 cpsi	96.2%	96.2%
EmiRec 600 cpsi	94.8%	94.8%
EmiRec 600 cpsi MOD	96.2%	96.2%

The five quantified uncertainties lead to the following evaluation function:

$$\eta_{th,rect}(T_{air}) = \alpha_{s,w} + \delta_{abs,rand} + \delta_{abs,dir} - \frac{\dot{q}_{loss,emi}(T_{air}) + \delta_{pos,emi} - \dot{q}_{refl,IR}}{I_{avg} + \delta_{FMAS} + \delta_{pos,sol}} \quad (7)$$

The uncertainties $\delta_{abs,rand}$, $\delta_{pos,emi}$, δ_{FMAS} and $\delta_{pos,sol}$ are all evaluated via method B of the GUM [25], i.e. they are assumed to be uniformly distributed uncertainties, while the directional uncertainty $\delta_{abs,dir}$ is instead evaluated via method A as detailed in Appendix A. Following the GUM, the uncertainty propagation is then given by:

$$u_{\eta_{th,rec}} = \sqrt{\sum_{\delta} \left(\frac{\partial \eta_{th,rec}}{\partial \delta} u_{\delta} \right)^2} \quad (8)$$

Here, variables u denote the standard deviation of each stochastic uncertainty variable δ resp. of the thermal efficiency. The results of the uncertainty propagation will be shown in Section 4.3 after the presentation of the efficiency results, because it has to be evaluated individually for each data set, i.e. for a specific air exit temperature level.

4. Results and discussion

4.1. Solar-weighted absorptivity

Needed for the evaluation of the absorber efficiencies, the solar-weighted absorptivity of each sample was determined with the methodology described in Section 3.2. The solar weighted absorptivity results for the different absorber samples are presented in Table 3. Three separate measurements were averaged for each absorber sample. In Table 3, the potential of the new 3D-shaped absorbers is already visible. The two StepRec samples absorb 6.1 percentage points more of the incoming irradiation than the reference HiTRec absorber, while the EmiRec designs absorb up to 5.1 percentage points more. During previous test campaigns, darkening of the metallic EmiRec absorbers caused by oxidation was observed. Because this proved beneficial w.r.t. the absorptivity, the manufacturer decided to forcibly oxidize the new absorber modules before this test campaign. The coarser 300 cpsi samples outperform the finer 600 cpsi channel density, because the increasing channel density (combined with the fixed wall strength) leads to a decreasing porosity. However, the additional retraction of every second corrugated layer allows the EmiRec 600 cpsi MOD design to reach the absorptivity level of the coarse absorbers.

4.2. Thermal efficiency

For each absorber and each temperature level of each test day, the taken infrared images were averaged. Then, the absorber positions within the infrared pictures were determined for each absorber set and each test day allowing the evaluation of the emission losses $\dot{q}_{loss,emi}$. Together with the average flux density I_{avg} and solar-weighted absorptivity values, the absorber efficiencies were calculated using Eqs. (1)–(4). The rectified thermal efficiency results are visualized in

Table 4

Rectified absorber efficiency interpolated for an air exit temperature of 650 °C for each absorber sample. For the HiTRec 80 cpsi absorbers, the results for the inner positions (6, 7, 10, 11) are shown. (D.S. = data set).

Absorber	Sample 1		Sample 2	
	D. S. 1	D. S. 2	D. S. 1	D. S. 2
HiTRec 80 cpsi	84.8%	85.3%	84.9%	84.6%
StepRec	91.7%	91.7%	91.5%	91.2%
EmiRec 300 cpsi	91.8%	91.6%	89.4%	90.1%
EmiRec 600 cpsi	90.6%	–	89.0%	89.3%
EmiRec 600 cpsi MOD	90.8%	91.0%	91.2%	–

Table 5

Uncertainty propagation for an air exit temperature of 650 °C and each absorber sample. All values are given as percentage points of the corresponding average flux density. (D.S. = data set).

Absorber	Sam.	D. S.	$\eta_{th,rect}$	$\eta_{th,rect}^{adj}$	$u_{\eta_{th,rect}}^{(1)}$	$u_{\eta_{th,rect}}^{(2)}$
HiTRec 80 cpsi	1	1	84.8%	83.9%	0.36%	0.68%
	2	1	85.3%	84.4%	0.32%	0.66%
	3	1	84.9%	84.0%	0.74%	0.94%
	4	1	84.6%	83.7%	0.47%	0.74%
StepRec	1	1	91.7%	91.2%	0.23%	0.43%
	1	2	91.7%	91.2%	0.25%	0.44%
	2	1	91.5%	91.0%	0.24%	0.43%
	2	2	91.2%	90.7%	0.26%	0.44%
EmiRec 300 cpsi	1	1	91.8%	89.3%	0.25%	1.61%
	1	2	91.6%	89.1%	0.25%	1.61%
	2	1	89.4%	86.9%	0.42%	1.65%
	2	2	90.1%	87.6%	0.31%	1.62%
EmiRec 600 cpsi	1	1	90.6%	88.2%	0.40%	1.59%
	2	1	89.0%	86.6%	0.50%	1.62%
	2	2	89.3%	86.9%	0.46%	1.61%
	1	1	90.8%	88.4%	0.29%	1.57%
EmiRec 600 cpsi MOD	1	2	91.0%	88.6%	0.28%	1.56%
	2	1	91.2%	88.8%	0.40%	1.59%

Fig. 6 over the air exit temperatures, while Table 4 lists the efficiency values interpolated for an air exit temperature of 650 °C.

Both Fig. 6 and Table 4 clearly show that all new absorber designs outperform the reference HiTRec 80 cpsi absorber. As has to be expected, the efficiency of all absorbers decreases with an increasing air exit temperature, because the emission losses increase. Still, the higher potential of the new designs holds for all evaluated air exit temperatures. In Table 4 interpolated efficiency values are given for an air exit temperature of 650 °C, since this is the target operating temperature for e.g. the Solar Tower Jülich power plant [21]. At this temperature level, the highest recorded efficiencies are 91.8% for the metallic EmiRec 300 cpsi and 91.7% for the ceramic StepRec design. This corresponds to an improvement of 6.5 resp. 6.4 percentage points over the HiTRec reference. The finer EmiRec absorbers also outperform the HiTRec design by at least five percentage points, while the differences between the three metallic designs are relatively small.

The ability of the new designs to outperform the HiTRec reference can be explained via a geometric comparison. The (average) porosity of each EmiRec design (86%–91%) is significantly higher than the 51% porosity of the HiTRec absorber, while the specific surface area is between two to four times as large as for the HiTRec absorber. The StepRec design also has a higher average porosity (78%) and a spec. surface area similar to HiTRec, but the step-pin front sections allow for a much deeper penetration of the solar irradiation. Overall, this enables the new designs to reach the measured high efficiency levels and outperform the state-of-the-art absorber.

4.3. Uncertainty analysis

The uncertainty analysis methodology which was described in Section 3.5 has been applied to the absorber efficiency results for an air

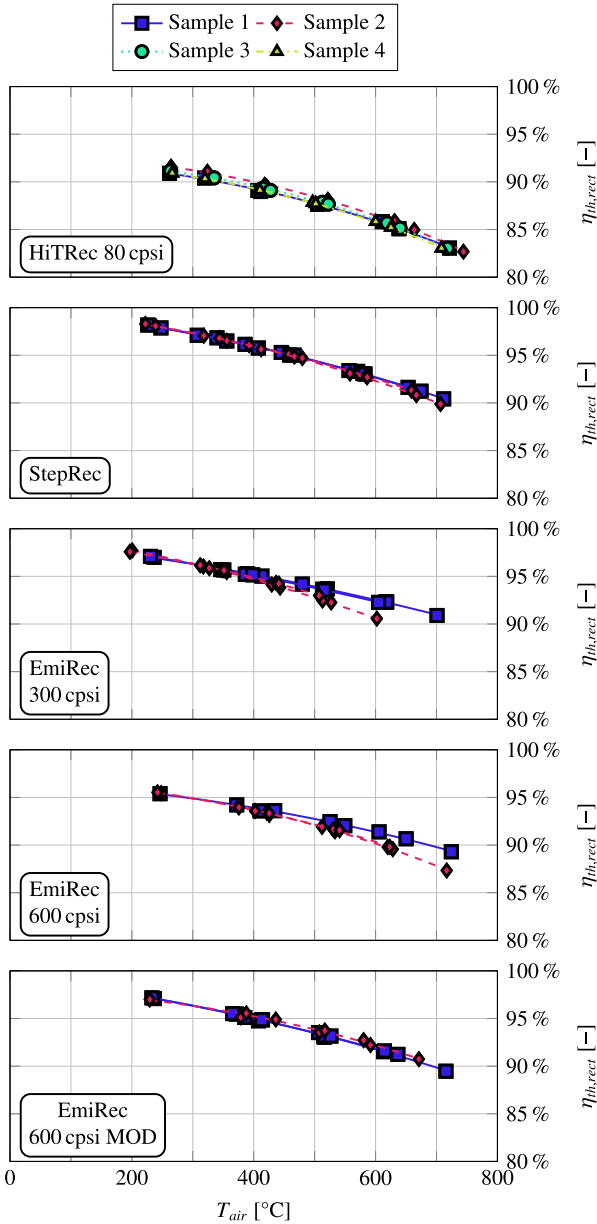


Fig. 6. Rectified absorber efficiency as a function of the air exit temperature for the five absorber geometries. Markers represent experimental data points, lines are regression functions based on the experimental data. For the HiTRec 80 cpsi design, the results for the samples placed on the inner positions (6, 7, 10, 11) are shown.

exit temperature of 650 °C. The results are listed in Table 5 and visualized in Fig. 7. The uncertainty propagation was first carried out without considering the irradiation direction uncertainty $\delta_{abs,dir}$ resulting in the total uncertainty for the absorber efficiency $u_{\eta_{th,rect}}^{(1)}$. The propagation calculation was then repeated while including the irradiation direction uncertainty, leading to the total uncertainty $u_{\eta_{th,rect}}^{(2)}$. The direction uncertainty was given this special treatment, because it differs from the other four considered sources of uncertainty. While the other four uncertainties are symmetric with expected values of zero, the direction uncertainty is one-sided for all absorbers with negative expected values (cf. Appendix A). For the second uncertainty propagation, the direction uncertainty was therefore transformed into a symmetric uncertainty by

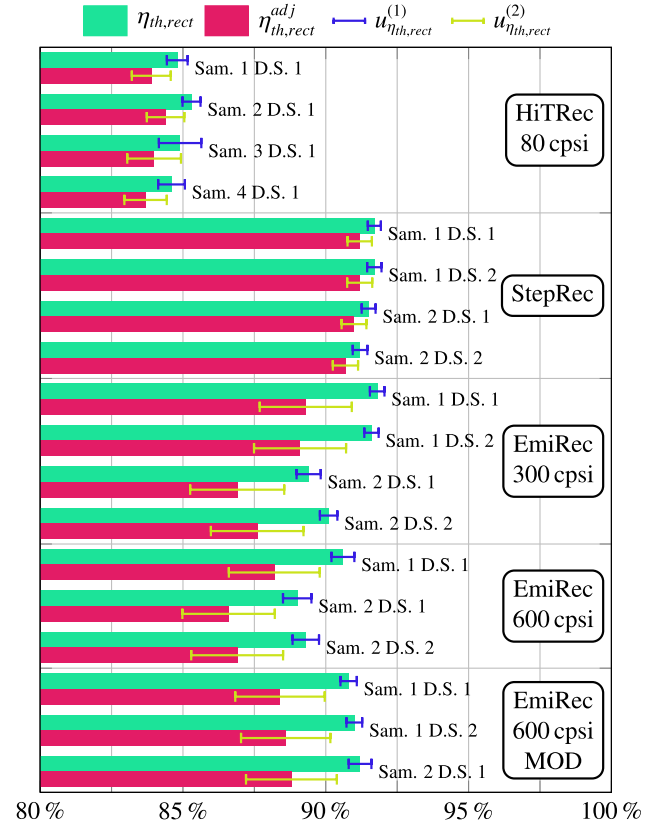


Fig. 7. Comparison of absorber efficiency and uncertainty for an air exit temperature of 650 °C and each absorber sample. All values are given as percentage points of the corresponding average flux density. (Sam. = sample, D.S. = data set).

defining an adjusted absorber efficiency:

$$\eta_{th,rect}^{adj} = \eta_{th,rect} - \mu_{\delta_{abs,dir}} \quad (9)$$

Here, $\mu_{\delta_{abs,dir}}$ is the expected value of the irradiation direction uncertainty. Relative to the adjusted efficiency, the direction uncertainty is then a normally distributed uncertainty with zero mean and the standard deviations as specified in Table A.2.

Without considering the direction uncertainty, the estimated standard deviation of the thermal efficiency is below one percentage point for all absorber geometries. The largest, individual source of uncertainty for this case is the general FMAS uncertainty δ_{FMAS} (cf. Section 3.5). However, the absorber efficiency is determined via a radiative energy balance, rather than a balance of the air enthalpy flow. This is beneficial for the uncertainty propagation of the FMAS uncertainty, because it only affects the emission loss fraction in Eq. (7), but not the solar-weighted absorptivity. Since the solar-weighted absorptivity is significantly larger than the (relative) emission losses, the effect of the FMAS uncertainty on the absorber efficiency is small.

The inclusion of the direction uncertainty leads to different results for evaluated absorber geometries. For the HiTRec and StepRec designs, the total uncertainty is still below one percentage point and the efficiency values are only reduced by 0.91 resp. 0.51 percentage points. In contrast, the efficiency uncertainty for the three EmiRec designs is increased to around 1.6 percentage points and the adjusted efficiencies are around 2.5 percentage points lower than the non-adjusted values. Standard deviations of less than two percentage points for all absorbers are considered good results for the experimental accuracy. The efficiency adjustments mean that the EmiRec absorbers now fall short of the efficiency level of the StepRec design, similar to numerical

results previously published in Broeske et al. [19]. But even with the adjustments, both new absorber types still outperform the reference HiTRec absorber by a clear margin of six (StepRec) resp. four (EmiRec) percentage points.

5. Conclusion and outlook

An experimental test campaign testing new 3D-shaped volumetric absorbers has been successfully conducted at DLR's high-flux solar simulator facility Synlight. The new designs included ceramic 3D screen printed absorbers manufactured by the company Exentis Technology GmbH and metallic foil absorbers from Vitesco Technologies Emitec GmbH. The new 3D-shaped porous absorber geometries showed the great potential for their use as part of open volumetric receivers: At air exit temperatures of 650 °C, (rectified) absorber efficiencies of up to 91.8% were recorded for the new designs. This represents an improvement of up to 6.5 percentage points compared to the state-of-the-art, ceramic HiTRec absorbers, which were also evaluated as reference. The absorber samples were tested for average flux densities of up to 650 $\frac{\text{kW}}{\text{m}^2}$ (second campaign) and air exit temperatures of up to 725 °C. The shown efficiency improvement of the new absorbers also extended to lower temperatures and the solar-weighted absorptivity, which was determined via auxiliary measurements. As the next step, a economical analysis of the new absorber designs will be conducted within the associated research project.

CRedit authorship contribution statement

Robin Tim Broeske: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Peter Schwarzbözl:** Conceptualization, Methodology, Resources, Writing – original draft, Supervision, Project administration, Funding acquisition. **Lisa Birkigt:** Investigation, Resources, Writing – original draft. **Srdan Vasic:** Resources, Supervision. **Sebastian Dung:** Investigation, Resources, Writing – original draft. **Till Doerbeck:** Methodology, Resources, Project administration. **Bernhard Hoffschmidt:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Estimation of the irradiation direction uncertainty

During the uncertainty quantification in Sections 3.5 and 4.3, systematic deviations of the irradiation direction(s) were considered as one source of uncertainty. The spectrometer, which was used to derive the solar-weighted absorptivity results, measures the sample reflectivity for a shallow incident angle of eight to ten degrees, while higher incident angles are possible for the Synlight facility. The resulting uncertainty $\delta_{abs,dir}$ was estimated via a comparison based on raytracing simulations, this approach is explained in the following.

A total of 84 lamps of the Synlight facility were used during the test campaign. While the positions of each lamp relative to the test rig's position are known, the locations of their focal points within the test rig aperture are unknown. Similarly, the flux distributions are known based on the FMAS measurements (cf. Section 3.4), but the individual contribution of a single lamp to each absorber position's flux level is not

available. Consequently, some simplifying assumptions are made for the uncertainty estimation. To start, the focal points of all lamps are assumed to be located at the center of the test rig aperture, which is given by the position vector p_{TR} . The (average) direction of the irradiation of the test rig which originates from lamp l is then calculated as:

$$d_l = p_{TR} - p_l \quad (A.1)$$

Here, p_l represents the position vector of lamp l . The tilt angle ti_l of each lamp is set as the angle between the average direction vector d_l and the normal vector of the test rig aperture n_{TR} :

$$ti_l = \arccos\left(\frac{d_l \cdot (-n_{TR})}{\|d_l\|_2 \cdot \|n_{TR}\|_2}\right) \quad (A.2)$$

The inverted aperture normal is used here, because it points into the test rig and as such corresponds to the main axis of the absorber channels. For the 84 lamps, the calculated tilt angles range from 7.8° to 53.8°. Similar to a histogram, the lamps are sorted into 11 groups based on their tilt angles. Each group j covers a width of five degrees and is represented by its average tilt angle ti_j , e.g. the third group contains all lamps with tilt angles between 10.0° and 15.0°, and is represented by the angle $ti_3 = 12.5^\circ$. A relative weight w_j is assigned to each group based on the number of lamps in the group n_j and the cosine of its representative angle:

$$w_j = \frac{\cos(ti_j) n_j}{\sum_{k=1}^{11} \cos(ti_k) n_k} \quad (A.3)$$

The cosine factor is used to account for the effect that lamps at a steeper tilt angle contribute less to the incident flux level compared to the same number of lamps with a shallower tilt angle. As a simplification, the solar-weighted absorptivity of each absorber is assumed to depend only the tilt angle, but not on the radial direction component. Based on the symmetry properties of the absorber structures, this assumption allows the histogram-like grouping of the lamps solely based on the tilt angle. Table A.1 lists the eleven tilt angle groups and their corresponding weights.

Using the raytracing software FEMRay [cf.27,28], the dependence of the solar-weighted absorptivity on the tilt angle was calculated for each absorber geometries. Per geometry, one raytracing simulation was carried out for each tilt angle group, except group one due to its zero weight. The irradiation was modeled as a cone with the average inclination set to the representative angle ti_j of the respective group. The opening angle of each lamp was estimated based on the radius of the lamp and the distance between the lamp and the test rig. This results in opening angles between 3.3° to 3.5° for all lamps, the opening angle of the irradiation cone was therefore set to 3.4° for all simulations.

Before the tilt angle evaluation, additional raytracing simulations were used to determine the surface/material absorptivity for each geometry. For a irradiation configuration meant to mimic the spectrometer measurements (parallel light at an inclination of ten degrees), the surface absorptivities were iteratively adjusted² until the simulated solar-weighted absorptivity matched the measured values (cf. Table 3).

In Fig. A.1, the simulated, solar-weighted absorptivity values are plotted over the tilt angle, while the spectrometer measurements are given as a reference. It can be observed that the StepRec absorber is least affected by the irradiation inclination due to its open step-pin design. Even for the highest, simulated tilt angle of 52.5°, the StepRec

² This lead to a surface absorptivity of 82% for HiTRec, 80% for StepRec and 60% for both EmiRec designs. For the EmiRec absorbers, the reflectivity was modeled as half specular and half diffusive, as this led to a closer agreement with measurements of 55% material absorptivity for the metal alloy. It should be noted that all three values potentially include implicit corrections for the uncertainties of the raytracing models, e.g. differences between absorber geometry model and the real sample geometry.

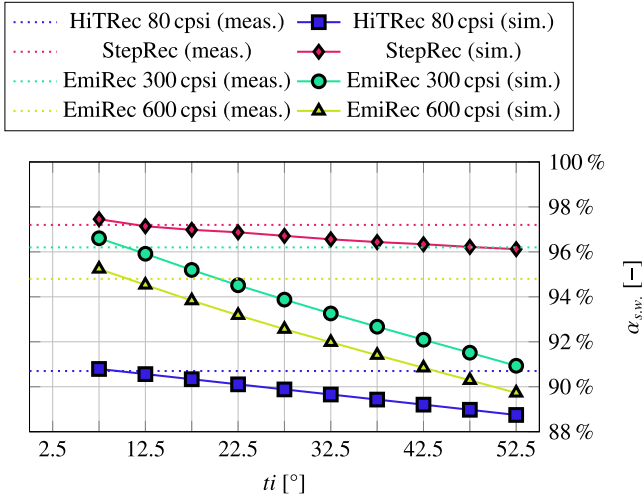


Fig. A.1. Solar-weighted absorptivity values as a function of the tilt angle. Solid lines represent raytracing simulation results, dotted lines the spectrometer measurements.

Table A.1
Discrete tilt angle distribution.

j	ti_j	n_j	$\cos(ti_j)$	w_j
1	2.5°	0	0.999	0.0%
2	7.5°	5	0.991	7.1%
3	12.5°	5	0.976	7.0%
4	17.5°	6	0.954	8.3%
5	22.5°	15	0.924	19.9%
6	27.5°	12	0.887	15.0%
7	32.5°	3	0.843	3.6%
8	37.5°	14	0.793	15.9%
9	42.5°	6	0.737	6.2%
10	47.5°	13	0.676	12.7%
11	52.5°	5	0.609	4.3%

absorptivity drops only 1.1 percentage point relative to the spectrometer measurement. The closed channels of the HiTRec absorber result in a stronger dependence on the tilt angle of up to two percentage points. Significantly stronger affected are the EmiRec absorbers with the much more finer absorber channels, the EmiRec 600 cpsi absorptivity drops up to 7.5 percentage points.

Based on the comparison of the simulation results to the spectrometer measurements, the irradiation direction uncertainty $\delta_{abs,dir}$ is modeled as a normally distributed uncertainty with the mean μ and the variance σ^2 given as:

$$\mu = \sum_{j=1}^{11} w_j (\alpha_{s.w.,sim.}(ti_j) - \alpha_{s.w.,meas.}) \quad (A.4)$$

$$\sigma^2 = \sum_{j=1}^{11} w_j (\alpha_{s.w.,sim.}(ti_j) - \alpha_{s.w.,meas.} - \mu)^2 \quad (A.5)$$

The resulting values for each of the four geometries are listed in Table A.2. The EmiRec 600 cpsi MOD design was not simulated, but is instead assigned the uncertainty of the similar EmiRec 600 cpsi variant. The effect of the irradiation direction uncertainty is discussed in Section 4.3.

Appendix B. Geometric absorber data

The HiTRec 80 cpsi absorber consists of simple square channels with a channel width of 2 mm, a wall thickness of 0.8 mm, and a depth of 40 mm to 50 mm. A square channel form is also the basis for the StepRec design with a constant wall strength of 0.44 mm and channel width of 2.29 mm. This design is however made up of five sections along the main

Table A.2

Mean and standard deviation of the irradiation direction uncertainty for each geometry.

Absorber	μ	σ
HiTRec 80 cpsi	−0.0091	0.0058
StepRec	−0.0051	0.0036
EmiRec 300 cpsi	−0.025	0.016
EmiRec 600 cpsi	−0.024	0.015

channel axis, within each section the geometric cross-section is constant (cf. part A of Fig. 2). Starting from the irradiated front, the five sections are:

- the first pin section: length 6.5 mm, relative width 25%
- the second pin section: length 6.5 mm, relative width 50%
- the third pin section: length 6.5 mm, relative width 75%
- the base channel: length 3.5 mm
- the shifted base channel: length 17.0 mm.

The width of the pins is defined relative to the total width of the StepRec absorber unit cell (2.73 mm, wall thickness + channel width). This means that, e.g., the each pin in the first section has the form of a plus-sign where both “lines” of the plus-sign are $0.25 \cdot 2.73 \text{ mm} = 0.6825 \text{ mm}$ long and 0.44 mm thick.

As described in Section 2.2, the metallic EmiRec absorbers are formed from alternating corrugated and smooth layers of thin sheets. For all three EmiRec variations, metal sheets with a fixed thickness of only 65 μm are used. The corrugated layers can be geometrically approximated as a wave function. For the 300 cpsi variant, the wave amplitude is 1.92 mm and the wave period 3.32 mm, while both 600 cpsi versions have an amplitude of 1.32 mm and a period of 2.38 mm. To increase porosity, the smooth sheet layers were shortened by 10 mm out of the total absorber depth of 40 mm (cf. part A of Fig. 3). In case of the EmiRec 600 cpsi MOD variant, every second corrugated layer was also shortened by 10 mm to further increase the porosity at the irradiated front.

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